

## Foundation (Jalapeno)

**Q1.** Factor the expression:  $x^2 + 5x + 4x + 20$

- (A)  $x(x + 5) + 4(x + 5)$
- (B)  $x(x + 4) + 5(x + 4)$
- (C)  $(x + 4)(x + 5)$
- (D)  $(x + 2)(x + 10)$
- (E)  $(x + 1)(x + 20)$

**Q2.** Factor the expression:  $3x^2 + 12x + 4x + 16$

- (A)  $3x(x + 4) + 4(x + 4)$
- (B)  $3x(x + 2) + 8(x + 2)$
- (C)  $(3x + 4)(x + 4)$
- (D)  $(3x + 2)(x + 8)$
- (E)  $3(x + 2)(x + 8)$

**Q3.** Factor the expression:  $2x^2 + 5x + x + 10$

- (A)  $x(2x + 5) + 2(2x + 5)$
- (B)  $x(2x + 1) + 10(2x + 1)$
- (C)  $(2x + 1)(x + 10)$
- (D)  $(2x + 5)(x + 2)$
- (E)  $(x + 1)(2x + 10)$

**Q4.** A space lab is launching a rocket with  $x^2 + 3x + 2x + 6$  kilograms of fuel.

- (A) The factored form of the expression is  $(x + 2)(x + 3)$
- (B) The factored form of the expression is  $(x + 1)(x + 6)$
- (C) The factored form of the expression is  $(x + 3)(x + 2)$
- (D) The factored form of the expression is  $(x + 6)(x + 1)$
- (E) The factored form of the expression is  $(x + 2)(x + 1)$

**Q5.** A physicist is analyzing data with the expression:  $4x^2 +$

$$12x + 3x + 9$$

- (A) The factored form of the expression is  $4(x + 3)(x + 3/4)$
- (B) The factored form of the expression is  $(4x + 3)(x + 3)$
- (C) The factored form of the expression is  $(4x + 9)(x + 1)$
- (D) The factored form of the expression is  $(2x + 3)(2x + 3)$
- (E) The factored form of the expression is  $(4x + 3)(x + 1)$

**Q6.** The mass of a planet is given by  $x^2 + 2x + 5x + 10$  kilograms.

- (A) The factored form of the expression is  $(x + 2)(x + 5)$
- (B) The factored form of the expression is  $(x + 5)(x + 2)$
- (C) The factored form of the expression is  $(x + 1)(x + 10)$
- (D) The factored form of the expression is  $(x + 10)(x + 1)$
- (E) The factored form of the expression is  $(x + 2)(x + 1)$

**Q7.** The number of moons orbiting a planet is given by  $3x^2 + 6x + 2x + 4$

- (A) The factored form of the expression is  $3(x + 2)(x + 1)$
- (B) The factored form of the expression is  $3(x + 1)(x + 2)$
- (C) The factored form of the expression is  $3(x + 1)(x + 4)$
- (D) The factored form of the expression is  $(3x + 2)(x + 2)$
- (E) The factored form of the expression is  $(3x + 4)(x + 1)$

**Q8.** The speed of a spaceship is given by  $2x^2 + 4x + 5x + 10$  meters per second.

- (A) The factored form of the expression is  $(2x + 5)(x + 2)$
- (B) The factored form of the expression is  $2(x + 2)(x + 5/2)$
- (C) The factored form of the expression is  $(2x + 2)(x + 5)$
- (D) The factored form of the expression is  $(x + 2)(2x + 10)$
- (E) The factored form of the expression is  $2(x + 5/2)(x + 2)$

### Open Ended Questions (FRQ)

**Q9.** Factor the expression:  $x^2 + 4x + 3x + 12$  and verify by multiplication

**Q10.** Factor the expression:  $2x^2 + 5x + 2x + 10$  and verify by multiplication

### Chef's Tips

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- Tip 1: Ensure students understand the importance of factoring by grouping in science applications
- Tip 2: Use visual aids to demonstrate factoring by grouping
- Tip 3: Encourage students to check their work by multiplying the factored expression